TORSION PRIMES OVER NUMBER FIELDS OF DEGREE ELEVEN

Audited computer-assisted proof, 28 October 2026

Theorem. Let S(11) be the set of primes p for which an elliptic curve over a number field of degree exactly 11 has a point of order p. Then

$$S(11) = \{2,3,5,7,11,13,17,19,23,29,31\}.$$

This document and work/audit11 replace the earlier proposed proof. The earlier nonic and decic exclusion results are not hypotheses of this proof. Their relevant local branches have been recomputed or replaced. The software dependencies are Magma, Sage, and the bundled modular-curve library; the mathematical dependencies are listed below.

1. OCCURRENCE AND THE EXACT-DEGREE REDUCTION

The primes 2,3,5,7 occur over Q and hence over a degree-11 extension. For each p in {11,13,17,19,23,29,31}, verify_occurrence.m constructs a nonsingular elliptic curve over a field of degree exactly 11, and checks that (0,0) is nonzero and is killed by p. All seven checks were rerun.

For p=11,13,19,23,29, respectively, the modular functions with exponent pairs (i,j)=(-6,-5),(-6,-1),(-2,1),(-1,1),(-1,0) on the standard (x,y) models have degree 11. The zero divisor of x^i*y^j-2 is one closed point of degree 11. For p=17, an element of L(11c), with c a rational cusp, and an irreducible specialization give the example. The script records the defining fields and Weierstrass coefficients, and verifies the points themselves. These assertions do not depend on a gonality search.

For completeness, the example at 31 uses a root a of the irreducible polynomial

$$f=t^{11}-t^{10}+2t^8+t^6-7t^5+t^4+4t^3-t^2+2t-1$$

and

$$y=(1036a^{10}-31a^9-503a^8+2108a^7+1676a^6+2654a^5 -5898a^4-4556a^3+2817a^2+2307a+3370)/2033,\ x=a,$$
$$r=(x^2*y-x*y+y-1)/(x^2*y-x),\ s=(x*y-y+1)/(x*y),$$
$$b=r*s*(r-1),\ c=s*(r-1).$$

On Y^2+(1-c)XY-bY=X^3-bX^2, the point (0,0) has order 31.

Conversely, for p>7, such a pair gives a noncuspidal closed point of degree 11 on X_1(p). Indeed, the residue degree divides 11, and degree 1 is excluded by Mazur’s theorem. We write D for its irreducible trace divisor of degree 11.

2. THE EFFECTIVE PRIME BOUND p <= 103

Oesterle’s bound, in the form proved in DKSS, gives p<=177989. The published strange-character table of Derickx--Stoll covers p<100000; it is not extrapolated beyond that range.

We first give the additional calculation above 100000. Let e be the winding element for X_0(p), T’_n=sum_{m|n} mu(n/m)T_m, and L_n=T’_(2n)-2T’_n. The integral spans of L_1,...,L_11 and of (T_2-3)T_1,...,(T_2-3)T_11 agree. DKSS’s intersection formula gives, for odd M, p>22M and pu=1 modulo M,

$$L_n e . \{0,a/M\} = \epsilon_M(na)-\epsilon_M(\nu a),\quad 1\le n\le 11,$$

where epsilon_M records which half of the nonzero residues contains its argument. We computed the resulting matrices for every unit u:

characteristic 3, M=53: 52 matrices of rank 11;
characteristic 5, M=73: 72 matrices of rank 11.

Since (T_2-3)Te lies in integral homology, this proves independence of T_1e,...,T_11e in Te/ell*Te. Parent's odd-characteristic criterion (Theorem 4.18) gives formal immersion of the winding-quotient map at 11 times the relevant rational cusp. Rational torsion on this quotient injects under reduction at the odd prime ell. Thus a degree-11 divisor which specializes entirely to that cusp is impossible. This step uses ell=3,5; no characteristic-2 optimal-winding-quotient criterion is used.

For p>100000 and ell in {3,5}, it follows that at some place of residue degree 1<=f<=11 at least one of the following conditions must hold:

(i) the Hasse interval for ell^f contains a multiple of p;
(ii) p divides ell^f-1 or ell^f+1.

Good reduction requires (i). Additive reduction cannot support such prime torsion: its identity component is additive and its component group has order at most 4. At multiplicative reduction, torsion in the torus requires (ii); otherwise the point is in the component group and the modular point has the common cusp type used above. On X_0(p) an Atkin--Lehner involution interchanges the two cusp conventions.

The tests at 3 and 5 exclude 6522 of the 6567 primes in 100000<p<=177989, leaving 45. At 3, good reduction for any surviving prime has f=11 and #E(F_(3^11))=p; the toric conditions fail. If p=1 modulo 3, the trace is divisible by 3, so E is supersingular and j=0. All four j=0 twists over F_(3^11) were enumerated afresh. Their group orders are 176419,177148,177877. None is one of the remaining primes congruent to 1 modulo 3. This removes 24 primes and leaves 21.

For each of those 21 primes, set m=(p-1)/2. The following prime powers r^v exactly divide m, with r different from 11. We verified every even character whose order is divisible by r^v to be non-strange:

p	r^v	orders	p	r^v	orders
176321	32	8	177473	32	4
176903	43	6	177533	44383	2
176921	4	4	177539	3061	2
176927	88463	1	177623	88811	1
176933	89	8	177647	12689	2
176951	25	2	177677	1033	4
177323	88661	1	177743	491	2
177347	359	4	177761	16	8
177383	2861	2	177791	773	4
177431	1613	4	177797	44449	2
177467	997	2			

All 73 tests were rerun. Over an exact auxiliary prime field, the local strange_primes.magma implementation removes the Eisenstein parts of the winding element and proves that its Hecke module has the full cuspidal dimension. A scalar Krylov sequence with linear complexity

equal to that dimension gives a rigorous rank lower bound. Reduction of the integral modular-symbol matrices then proves non-strangeness in characteristic zero. Random choices can fail to find such a certificate, but cannot create a false full-dimension certificate. Galois conjugacy covers all characters of each tested order.

Let str(p) be the least common multiple of the orders of the strange characters, with str(p)=1 when there are none. The selected tests show v_r(str(p))<v. Thus r divides m/str(p), and m/str(p) cannot divide 11. Derickx--Stoll Corollary 7.5 excludes these 21 primes: its small-prime alternative fails, its two exceptional alternatives require even degree, and its divisibility alternative has just been excluded.

For 137<=p<100000, the same corollary and the published Table 2 exclude all 9560 primes. Specifically p^2>(2^19/325)*11+1, and m/str(p) does not divide 11. The application is to S_new(11), which is legitimate by the exact-degree argument in Section 1.

For p=107,109,113,127,131, the prime is non-strange. The gonality bound 325(p^2-1)/65536 is greater than 55. Derickx--Stoll Proposition 4.1 with diamond <2>, ell=2 and n=5 makes D a union of diamond orbits. Remark 4.4 supplies its rational 2-primary torsion hypothesis. The orders of <2> are 53,18,14,7,65. Neither these lengths nor their possible halves or thirds divide 11. Irreducibility of D gives a contradiction.

The remaining primes to exclude are therefore

37,41,43,47,53,59,61,67,71,73,79,83,89,97,101,103.

3. RATIONAL-CUSP REDUCTIONS AT 2

For each remaining prime we recomputed the DKSS endomorphism version of the formal immersion criterion. This uses actual endomorphisms t=t_1(T_n)*(T_q-<q>-q) on intermediate Jacobians. The winding calculation proves that t_1 has finite rational image, and the odd Hecke factor kills rational torsion. Hence t annihilates the rational Jacobian group. Independence of the prescribed operators modulo 2 proves formal immersion without an assertion about an optimal quotient’s cotangent lattice at 2. At 41,61,67 we use products of several such maps.

Every normalized cusp multiplicity tuple (n_0,...,n_(i-1)) of total 11 was checked, with n_0 maximal. All other entries are at most 5. The diamond indices i and exact numbers of checked tuples are:

p	i	tuples	p	i	tuples
37	18	1921834	71	5	317
41	20	4525972	73	6	872
43	21	6768091	79	13	152026
47	23	14476848	83	41	2580289822
53	13	152026	89	4	102
59	29	107154192	97	4	102
61	15	454226	101	5	317
67	11	44444	103	17	1216031

All 2717157222 checks passed. The row compression is verified to be injective on the span of the original operators. A separate dynamic program gives exactly the same tuple count in every case, so a search pruning rule cannot silently omit a configuration in a successful run. For the combined maps, independence after the recorded joint linear

projection implies independence of the unprojected operators.

These checks apply after pushing D to the intermediate curve. Rational cusps on X_1(p) map to rational cusps there. Formal immersion and the annihilation of rational Jacobian points force a divisor with such a reduction to be its cuspidal lift, a contradiction.

4. A COMPLETE FRESH FINITE-FIELD CENSUS

The script census_sources.m enumerates all j-invariants and twists over F_(2^d), for every 1<=d<=11 and all 16 remaining primes. It verifies p^2 does not divide #E(F_(2^d)) whenever p divides it. Consequently the rational p-torsion is cyclic, and multiplying one generator by every nonzero scalar gives every relevant level structure. The Tate (x,y) coordinates determine the moduli point and its exact residue degree. Frobenius-invariant keys remove repetitions and enumerate full closed diamond orbits, including points whose j-invariant lies in a subfield. This is a census of actual fine-moduli points, not only j-invariants.

There are 210 diamond orbits of noncuspidal closed points of degree at most 11 in total. Of these, 206 are ordinary and four supersingular. The ordinary calculation is summarized below. The column ’excluded’ counts fresh full-rank principal-divisor obstruction matrices.

p	genus	ordinary orbits	excluded	section power
37	40	25	21	2
41	51	23	23	2
43	57	17	17	2
47	70	15	15	2
53	92	13	13	1
59	117	15	15	1
61	126	11	11	1
67	155	16	16	1
71	176	15	15	1
73	187	7	7	1
79	222	5	5	1
83	247	14	14	1
89	287	7	7	1
97	345	5	5	1
101	376	8	8	1
103	392	10	10	1

All 202 exclusion matrices were independently checked using Sage/M4RI. The four ordinary exceptions at 37 are treated in Section 7. The four supersingular orbits are at (p,d)=(37,9),(41,5),(43,7),(73,6).

The minimum noncuspidal degree is at least 6 except at 41. At 41 the minimum degree is 5, all degree-5 points are supersingular, there are no degree-6 or degree-7 points, and the minimum ordinary degree is 8. Consequently a degree-11 reduction has at most one noncuspidal closed point, except for a possible pair of degree-5 points at 41. In particular a mixed degree-5 plus degree-6 support cannot occur. Its complementary cusps are rational: every nonrational cusp has degree too large to coexist with a noncuspidal point of these minimum degrees.

5. THE RECOMPUTED ORDINARY HECKE OBSTRUCTIONS

Here diamonds act on pairs by delta_a(E,P)=(E,aP). Use the geometric 3-isogeny correspondence and put A=T_3-1-3delta_3. In this convention

A kills rational torsion and kills each rational cusp divisor. This is the inverse-diamond convention to the modular-form version used in Section 3; the Tate-curve check records the conversion explicitly.

Let sigma denote the diamond on the canonical coordinate column whose matrix is exported with the q-expansion basis. It acts on the moduli pair as delta_r^(-1), where r is the recorded prime primitive root. We use the following endomorphisms alpha:

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p=41,47,59,71: alpha=1;
p=61:          alpha=sigma^6-1;
p=97:          alpha=(sigma-1)*(sigma^4-sigma^2+1);
p=101:         alpha=(sigma-1)*(sigma^5+1);
all others:    alpha=sigma-1.
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In every case alpha J_1(p)(Q) is finite. At 41,47,59,71 this is the rank-zero result in DKSS. At the other non-strange primes alpha kills the trivial-character part. For 61,97,101 the only strange character orders are respectively 6,12,10. The indicated polynomials kill those characters as well as the trivial character. Kato’s theorem applied to the other newform factors gives the required finiteness. Inverting diamonds does not change any of these character-order assertions.

If the reduction of D is P+C, with C rational cuspidal, then alpha*A*P must be principal: alpha*[D-11c] is rational torsion, A kills it, and A*C=A*c=0. This argument is independent of the degree of P. It is why all lower-degree branches must be checked, even though D has degree 11.

The computations proving non-principality use the full modular curve, not an auxiliary low-genus quotient. For each p, a saturated integral basis of weight-2 forms supplies the canonical coordinates modulo 2. The modular units

$$x=(s-r)/(rs-2r+1),\quad y=(rs-2r+1)/(s^2-s-r+1)$$

recover coordinates at every ordinary source and Hecke target used. All unit relations are certified globally: their zero orders at a rational cusp exceed 2g-2 plus the relevant pole degree. The pole degrees were recomputed independently at every cusp using Tate curves; at the nonrational cusp type the two functions are units. They are modular units because their factors are the low-order modular equations. This prevents the plane models’ bidegrees from being an unverified assumption in the zero-order argument.

The q-expansion model is transported to the point-level curve by the Fricke isomorphism from the multiplicative-level model. The Tate pair (E(q^p),q) maps to E(q) with its canonical multiplicative level structure. Thus the transported differentials have expansions f_i(q)dq/q at this rational cusp; the level normalization is a 2-adic unit. Fricke extends over Z_2 since p is odd.

For clarity, finite-truncation saturation already gives the full integral differential lattice at 2. Let L=H^0(X/Z_2,Omega), and let B be the stored q-precision. In every case B-1>2g-2. If a differential in L tensor Q_2 has integral first B coefficients but is not in L, multiply it by the smallest power 2^a putting it in L. Its reduction is nonzero, while its first B coefficients vanish modulo 2. It therefore vanishes at this rational cusp to order at least B-1>2g-2, impossible for a nonzero holomorphic differential. Hence integral truncations correspond exactly

to L, and the saturated rational q-basis, after tensoring with Z_2, is the full lattice. Smooth proper base change then gives every differential on X_F2.

The Fricke description also explains the inverse action of point diamonds. The diamond convention is checked globally using the Tate point rP. At each ordinary point the combined unit-relation matrix has rank g-1, so its projective kernel is the required canonical point. Eight coefficients of local jets are computed by Hensel lifting the p-division polynomial and the elliptic-curve equation. The parameter varies a6 in y^2+xy=x^3+a2*x^2+a6; here j=1/a6, so it is a uniformizer on the ordinary locus. The derivative and all jet equations are checked.

The 3-division polynomial has four distinct roots, one for each cyclic 3-subgroup up to Frobenius. Factoring it and applying the isogeny with kernel polynomial X-x_3 gives every target directly. If a kernel factor has degree e over a source field of degree d and its target moduli point has residue degree t, its trace weight is de/t. All weights are checked and the resulting Hecke divisor has degree 4d. Proper j-fields, repeated targets, support cancellations and multiplicities are retained.

Here is the exact linear-algebra obstruction. Write alpha*A*P=U-V, with U,V effective and disjoint, and let R=H^0(K_X^k), with k as in the table. Choose a nonzero F in R vanishing on U. If U-V=div(g), the nonzero section S=F/g vanishes on V, and H*S belongs to F*R for every H in R vanishing on U. The recorded matrices impose these necessary linear conditions on S. Full rank equal to dim(R) for the stored row matrices leaves only S=0, a contradiction. Every multiplication tensor identity is checked by q-expansions beyond its Sturm bound. The independent rank checks certify the 202 listed non-principality assertions.

The native models at 37,41,43 have an additional identification check: their plane equations vanish on the Tate-unit series to orders 2365,2358,2355, respectively, exceeding pole bounds 684,1012,1248. Their degrees in y equal the independently computed degrees of x, namely 18,22,24. Since x,y generate the modular function field, these are the required modular-curve models, not extraneous components.

6. NONRATIONAL CUSPS AND THE SUPERSINGULAR BRANCHES

The only nonrational cusps of degree at most 11 are:

p=41: degree 10; p=43: degree 7;
p=73: degree 9; p=89: degree 11.

At 41 there are two degree-10 cusps. The class A applied to either is represented, up to sign, by twice their difference. It was recomputed to be non-principal. The two degree-5 supersingular points are exchanged by a diamond. A applied to one of them is non-principal, and A applied to each of the three unordered pairs, including repetition, is non-principal. Two calculations suffice for the pairs by diamond symmetry. All these native Riemann--Roch checks were rerun.

At 43 the three degree-7 supersingular points form one diamond orbit. Their alpha*A class is non-principal by a fresh native calculation.

We use the following stabilizer argument for the remaining branches. Suppose h fixes the noncuspidal or nonrational-cusp part of red_2(D), and C is the rational lift of its rational-cusp complement, of degree e. Put Z=D-C and y=[hZ-Z], a degree-zero class in J(Q). Although Z

need not have degree zero, y is torsion if (h-1)J(Q) is finite. Indeed, let m be the order of h and put N_h Z=sum_(i=0)^(m-1) h^i Z. The divisor mZ-N_h Z has degree zero, and

$$m\,y = (h-1)[mZ-N_h\,Z].$$

The right side is torsion, so y is torsion as well. Also red_2(y)=0. The kernel of reduction on rational torsion at a prime of good reduction 2 has exponent at most 2, so 2y=0. If J(Q)[2]=0, then y=0. Alternatively, if h has odd order m and acts trivially on J(Q)[2], then the norm relation gives 0=(1+h+...+h^(m-1))y=my, so again y=0. If gon_Q(X)>11+e, linear equivalence now forces

$$hD+C = D+hC.$$

Cuspidal and noncuspidal supports are disjoint. Thus hC=C. The action on rational cusps is free, so m must divide e.

The rational 2-torsion assertions used here were recomputed on the actual integral homology of JOne, sign 0, rather than on an isogenous Jacobian. The common kernels modulo 2 of A_3 and A_5 have dimensions 0,6,2,0,6 for p=37,43,67,71,73. On them the tested diamonds <2>,<2>,<38>,<32>,<2> act trivially. The integer diamond matrices come from (T_q^2-T_(q^2))/q before reduction. Sage independently verifies these kernels and actions. Only 37,43,73 are needed in this final proof.

At 43, h=<2> has order 7 and fixes each degree-7 closed cusp. The complement has degree e=4. Section 8 proves gon_Q(X_1(43))>=16>15, so the stabilizer argument would imply 7|4.

At 73, h=<2> has order 9. It fixes each degree-9 closed cusp, and the degree-6 supersingular closed points. The complements have degrees 2 and 5 respectively. The gonality bound is greater than 26. The same argument would imply 9|2 or 9|5. The stabilizer of the supersingular points is checked directly on their fresh Tate-coordinate records.

At 89, a nonrational cuspidal reduction is a single degree-11 cusp, fixed by h=<2> of order 11. The class y=[hD-D] is torsion and 2y=0. Here gon_Q(X_1(89))>39>22, so the principal divisor 2hD-2D forces hD=D without a 2-torsion action calculation. An order-11 diamond has no noncuspidal fixed points in characteristic zero. Hence D is one h-orbit, and its image on X_0(89) is a noncuspidal rational point, contrary to Mazur’s prime-isogeny theorem.

7. THE EXCEPTIONS AT 37

The published Q-gonality of X_1(37) is 18 (Derickx--van Hoeij). The fresh ordinary sieve leaves exactly these four diamond orbits:

source index	degree	closed diamond orbit size
1	6	3
5	9	2
12	10	18
13	10	9

There is also the supersingular degree-9 orbit, of size 1. Since J_1(37)(Q)[2]=0, the stabilizer class in Section 6 vanishes directly.

For degree 6, h=<8> has order 6 and fixes the closed points. The cusp

complement has degree 5, and 18>11+5. Thus 6 would divide 5. This replaces the earlier argument using known sextic characteristic-zero lifts and 26334 cotangent restrictions; neither is a dependency here.

For both degree-9 survivors, including the supersingular orbit, h=<4> has order 9, and 18>11+2. This would require 9|2. For the degree-10 orbit of size 9, h=<6> has order 2, and 18>11+1. This would require 2|1. Each claimed stabilizer is verified on the fresh records.

The remaining orbit is the reduction of an actual degree-10 divisor S. Its characteristic-zero witness uses

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f=t^10-5t^8+3t^7+10t^6-8t^5-7t^4+9t^3-3t+1,
g=2t^9+t^8-10t^7+t^6+23t^5-6t^4-22t^3+11t^2+9t-5,
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with x a root of f, y=g(x), and the same r,s,b,c formulas as in Section 1. Irreducibility, nonsingularity and order 37 were rechecked. Its reduction was matched directly to source orbit 12.

Put alpha=<2>-1. On the native smooth model over F_2 we computed its cotangent image, of dimension 38, and restricted it at S plus each of the 18 rational cusps. Every restriction has rank 11. These 18 ranks, and the global diamond/cotangent calculation, were independently replayed. The reduction of the putative D is a diamond translate of red(S)+c for one of these cusps; all other translates follow by commutation with diamonds. The two alpha images agree after reduction, and hence agree in characteristic zero since rational torsion has injective reduction here. Formal immersion at the common reduction forces D to be that degree-10 divisor plus a cusp, contradicting irreducibility of D. This excludes the last branch at 37.

8. A FRESH GONALITY CERTIFICATE AT 43

Only the bound gon_Q(X_1(43))>=16 is needed. The weaker bound >=13 quoted in the published gonality paper is insufficient for Section 6; the stronger website computation is therefore replaced here.

Let X=X_1(43) over F_2, with genus 57. It has exactly 21 rational points, all rational cusps, no closed points of degrees 2,...,6, six of degree 7, and 42 of degree 8. The latter consist of two diamond orbits of size 21 distinguished by their j-polynomials. These assertions are checked by native place computations and the independent elliptic-curve census.

If a function has degree at most 15, one of its three rational fibers contains at least seven rational points. After a Mobius transformation over F_2 we may take this to be its pole fiber. There can be at most one nonrational pole. Its degree is 7 or 8: degree 9 together with seven rational poles already exceeds 15. For a degree-7 pole P, the rational poles have total degree at most 8, so at most one is double. Diamond translation moves a possible double pole to a fixed rational cusp P_1. Native Riemann--Roch gives

$$l(C+P_1+P)=1 \text{ for all six degree-7 } P,$$

where C is the sum of the 21 rational cusps. For a degree-8 pole Q, all rational poles are simple, and

$$l(C+Q)=1$$

for a representative of each of the two full diamond orbits. Since C is diamond invariant, these eight native calculations exclude every case with a nonrational pole.

For rational poles only, pad their multiplicities to total 15 using an existing pole. There are 66 partitions of 15 of length at least 7. If all multiplicities are at most 3, choose an unused rational cusp and translate it to P_1; use the envelope 3(C-P_1). Otherwise translate a largest pole to P_1. The partitions are covered by the following envelopes (P,Q rational cusps):

at most two multiplicities >1:	C+8P_1+4P;
one multiplicity >2, after that:	2C+5P_1;
two multiplicities >2:	2C+4P_1+2P;
three multiplicities >2:	2C+3P_1+2P+2Q.

In the last line P,Q are distinct and different from P_1. Four multiplicities >2 with one >3 are impossible. After the first case is removed, the maximal multiplicities in the next three cases are 7,6,5; the other large multiplicities are at most 4. These inequalities and the coverage of all 66 partitions are checked explicitly.

We evaluate a basis of holomorphic differentials at all rational cusps to precisions 13 at P_1 and 5 at the others, using the certified full q-expansion basis and diamond transport. For a differential sum a_n q^(n-1)dq, these are the coefficients a_1,...,a_cap. The resulting 57 by 113 matrix has rank 57. By the residue pairing, the kernel of its columns allowed by an envelope is exactly the space of realizable principal parts of functions, modulo constants. For a vector of principal parts, the pole degree is the sum, over the cusps, of the largest index of a nonzero pole coefficient.

All 234 envelope instances are checked. Their principal-part kernels have dimension at most 4. Every nonzero vector in every kernel has pole degree greater than 15. Sage separately checks the kernel dimensions and every vector equation. As a native control, the first envelope has dimension 4 after normalizing the function’s value at P_1, and the same 15 pole degrees in both calculations:

30,30,36,36,36,36,39,39,39,45,45,48,48,51,54.

This proves gon_F2(X)>=16. Gonality cannot increase under good reduction, so gon_Q(X_1(43))>=16 as required.

9. AUDIT SCOPE AND REPRODUCIBILITY

The following calculations were performed afresh in this audit: the occurrence witnesses; all 73 selected character tests; both odd formal immersion matrix families; the prime-bound arithmetic; all 16 rational-cusp exports and exhaustive checks; the complete point census through degree 11; all 16 full canonical/unit constructions; all ordinary 3-isogeny targets and jets; all 202 principal-divisor matrices and independent ranks; the native exceptional checks at 41 and 43; the integral-homology 2-torsion bounds; the 18 decic-plus-cusp restrictions at 37; and the gonality certificate at 43.

The earlier read-only replay scripts at 37,67,71,73,79,83,89 were also run successfully. Their old quotient exclusion results, and the old

97/101 lifting and 103 quotient-norm calculations, are not needed in the final argument: the fresh full-curve obstructions replace them. Intermediate unsuccessful computations are excluded from the final certificate archive. Old fixed log labels mentioning 83 in generic unit routines do not specify the level; the serialized level, genus, dimensions and final AUDIT markers do.

The final case ledger is work/audit11/final_result.json. The archive contains the proof, final source files, data and logs, the modular-curve library and its license, and a SHA256 manifest. The read-only closure checker verifies the case ledger against these certificates; it is not itself a replacement for the mathematical proof above. The replay driver also regenerates the native and linear-algebra calculations. Exact reproducibility uses Magma V2.30-1 and Sage’s Python.

All possible reductions for every remaining prime have now been excluded. Together with Sections 1 and 2, this proves the theorem.

MATHEMATICAL REFERENCES

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https://arxiv.org/html/2505.14109v1

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https://arxiv.org/abs/1307.5719

Mazur’s torsion and prime-isogeny theorems; Kato’s rank-zero theorem for newform factors with nonzero central L-value; standard good-reduction, formal-group and gonality-specialization results.